Exercises Hydrological Modelling

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1 Introduction

This book contains exercises for the course Hydrological Modelling at Ghent University. The focus of this set of exercises will be on rainfall-runoff modelling with a conceptual hydrological model.

The objective of the practical exercises of this course is to illustrate some of the most important steps necessary to perform a proper modelling of a medium-scale catchment in Flanders. Only some specific steps of the process will be illustrated here across 5 exercises:

1. Delineation of the catchment (Chapter 4)
2. Implementation of a hydrological model (Chapter 5)
3. Calibration of the hydrological model (Chapter 6)
4. The analysis of uncertainty in the model output of the model (Chapter 7)
5. Assimilation of observations into the model (Chapter 8).

Needless to say that for each step, there are many options, and that the choice for e.g. a specific model, calibration algorithm, or evaluation criterium depends on the specific situation, the aim of the modelling exercise, and the personal taste of the modeller. For an extensive – yet non-exhaustive – overview of different techniques in hydrological modelling, we refer to the theoretical course notes and specialized literature.

The aim of the practicals is to:

1. Make you acquainted with the advantages and disadvantages of the selected algorithms,
2. Make you aware of the potential problems related to modelling rainfall-runoff relationships, and
3. Point you to and make you reflect on existing alternatives for the algorithms selected here.

2 Practical information

Before diving into the exercises, some practical information is provided on how to effectively set up your working environment.

2.1 The repository

All code used to create this assignment is available at https://github.com/olivierbonte/hydrological-modelling. This collection of files is called a repository. To get started, go the the URL above, click on the green `Code` button and select `Download ZIP` (see Figure 2.1). Unzip the downloaded file and place the resulting folder at a location of your choice. This folder contains all the files you need to complete the exercises.

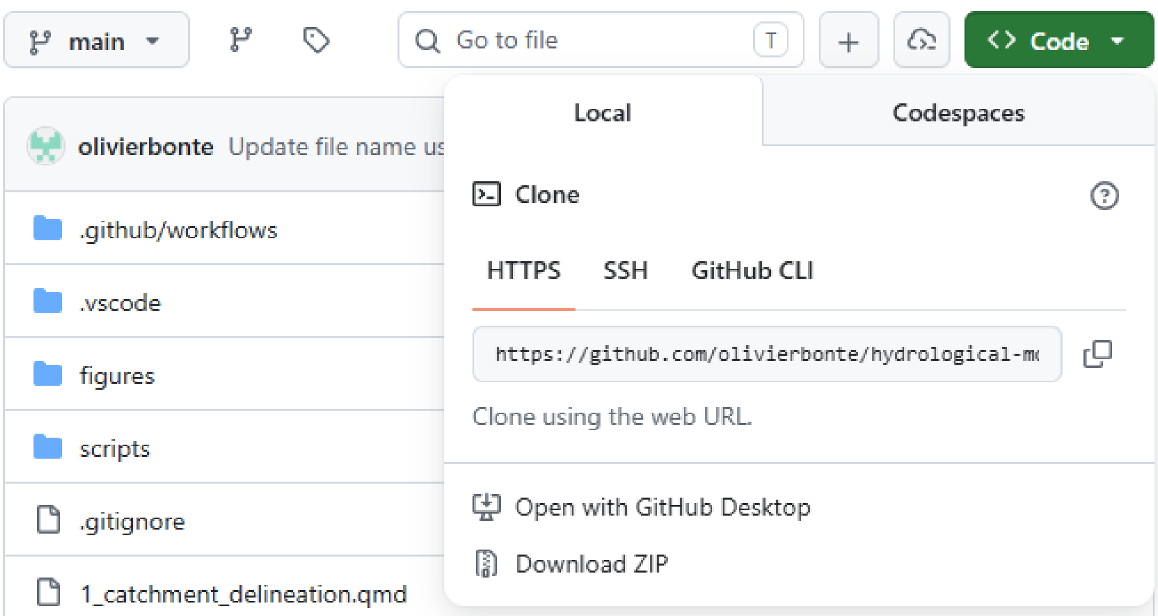

Figure 2.1: Screenshot of GitHub download button

The data needed is not included in the repository, and should be downloaded separately (see Chapter 3 for more information).

Note that this assignment is provided to you in two formats, up to you which one you prefer to work with:

1. A website at https://olivierbonte.github.io/hydrological-modelling/
2. A PDF document, which you can download from https://olivierbonte.github.io/hydrological-modelling/Exercises-Hydrological-Modelling.pdf

2.2 Working with Quarto and Python

The idea of this exercise set is that you will use Quarto to combine your code and your scientific report in a single document.

2.3 Evaluation

3 Study area and data

For these exercises, a hydrological dataset of the Zwalm catchment will be used. The Zwalm catchment is located in the center of Belgium and can be considered a medium-scale catchment. Its main tributary, the Zwalmbeek, is visualised in Figure 3.1. During the period 2009-2025, a dataset of daily precipitation, potential evapotranspiration and discharge is at your disposal. The average annual rainfall and potential evapotranspiration of the catchment are 735 mm and 561 mm respectively, typical values for Belgium.

All the data needed for this exercise is available on Ufora as `data.zip`. Download this file, unzip it and place the resulting `data` folder in the root of this repository (that means at the same level as e.g. `scripts`).

Alternatively, you can download and process the data yourself by following the instructions in `scripts/data_download/README.md` and `scripts/data_process/README.md`.

3.1 Catchment data

3.2 Meteorological and discharge data

3.3 Digital terrain model

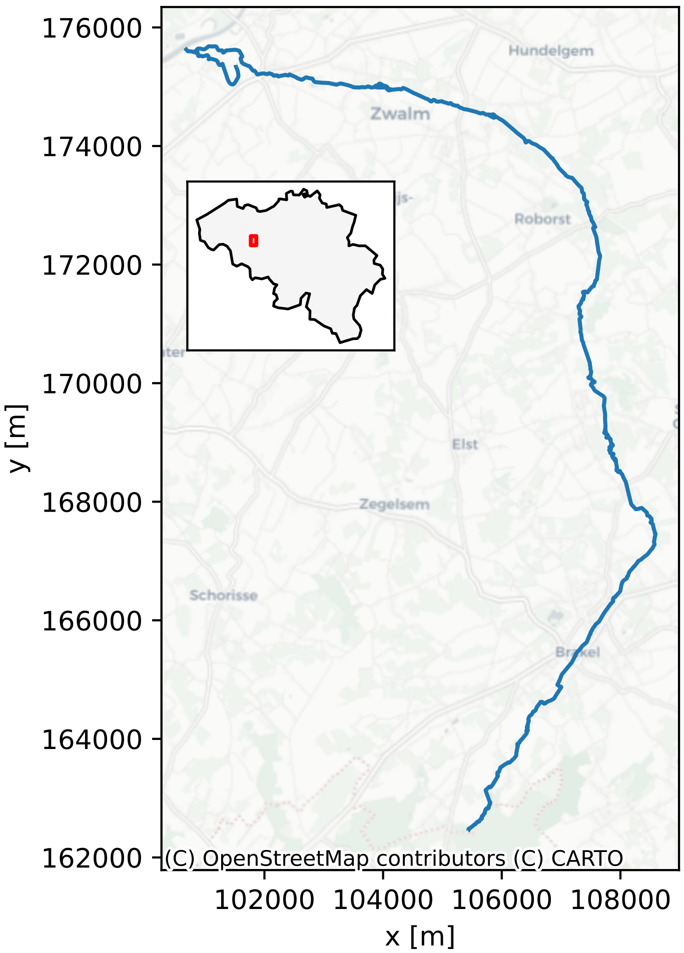

Figure 3.1: The Zwalmbeek and its location within Belgium (CRS: EPSG:31370).

4 Exercise 1: Delineation of the Zwalm catchment

5 Exercise 2: Implementation of the model

6 Exercise 3: Calibration of the model

7 Exercise 4:Uncertainty analysis of the model output

8 Exercise 5: Data assimilation

References